

Projects in AI and ML -- Spring 2026

Homework 6

This is the last homework of the semester and is due on April 13, 2026 at 11:59 pm.

General submission requirements

- Submit all code and writeup in one or more notebooks if needed. Your submission should be organized and easy to run.
- Include plots, tables, or screenshots where appropriate.
- Cite all external resources, datasets, code repositories, and papers you use.

Part 1: Reinforcement Learning

Task 1 (40 points): Implement value iteration or policy iteration for a small discrete Markov Decision Process (MDP).

You may use a grid world, FrozenLake, inventory control, or another small discrete environment approved in class.

Perform the following:

- Clearly define the state space, action space, reward structure, and discount factor.
- Implement either value iteration or policy iteration and compute the final policy.
- Show the learned value function and the final policy in a readable form.
- Run at least one small experiment showing how the policy changes when you vary either the discount factor or the reward design.
- Briefly discuss why this setup is an MDP and what the learned policy is doing.

Part 2: Mini Research Task on a Novel AI Topic

Task 2 (60 points): Complete one scoped mini research task on a recent AI topic. Diffusion models are strongly encouraged, but you may choose another novel topic.

Examples of acceptable topics include diffusion models, consistency models, flow matching, multimodal foundation models, retrieval-augmented generation, parameter-efficient fine-tuning, graph neural networks, or another recent topic in the field of AI.

Perform the following:

- Select one focused topic and identify at least two technical sources. At least one source should be a primary source such as a research paper, official documentation, or a model card.
- Explain the problem the method is trying to solve, the core idea behind the method, and the main architecture or training objective in your own words.

- Include one small hands-on component. For example, you may run a simple demo, inspect a pretrained model or pipeline, reproduce one small figure or result, compare two variants, or analyze how one key design choice affects outputs.
- If you choose diffusion models, possible hands-on directions include comparing different numbers of denoising steps, comparing schedulers, testing prompt sensitivity, or explaining the forward and reverse diffusion process with a compact experiment.
- Include at least one figure, table, or carefully labeled output example that supports your analysis.
- Discuss at least two limitations, risks, or open challenges related to the method.
- Propose one concrete extension, improvement, or follow-up experiment that you would pursue next.

The goal is depth on one focused recent topic, not breadth across many topics.

Task 3 is for 6000 level only

Task 3 (100 points): Select one recent AI/ML research paper from 2024 to 2026 that is related to your chosen novel topic or another advanced course topic that interests you.

Prepare a technical review and a small reproducibility component.

- Write a 2-to-4-page review (**submit a pdf or a link to the pdf**) summarizing the problem, method, dataset, evaluation metrics, and main findings.
- Identify at least two strengths, two limitations, and one reproducibility challenge in the paper.
- Reproduce one figure, table, ablation, or small experiment from the paper. If full reproduction is not feasible, perform a structured reproducibility audit and explain exactly what blocked reproduction.
- Suggest one concrete extension or follow-up experiment that would strengthen the paper.
- Submit your review together with any code, notebook, and references used.

Clear reasoning, reproducibility, and presentation quality matter.